AIGR TreeGrafter Function-Inference Stress Test — Final Report

Target: *Pseudomonas putida* KT2440 RelA (UniProt **Q88MB8**, gene `relA`, organism code PSEPK, NCBITaxon:160488) **Seed hypothesis under test:** relA has **GTP diphosphokinase activity (GO:0008728)** — propagated as an IEA via TreeGrafter/PANTHER, original reference GO_REF:0000118. **Verdict: SUPPORTED — retain GO:0008728.**

Summary

The automated phylogenetic pipeline (TreeGrafter/PANTHER) propagated **GO:0008728 (GTP diphosphokinase activity)** to *Pseudomonas putida* KT2440 RelA (Q88MB8). Our independent, primary-evidence evaluation finds this propagation **correct**. The protein is a canonical **"long" RelA/SpoT homolog (RSH)** of the RelA synthetase clade: its central (p)ppGpp-synthetase / nucleotidyltransferase domain is **58.7% identical** to the experimentally validated *Escherichia coli* RelA synthetase, and the metal-coordinating catalytic **FDDY** motif is fully intact. GO:0008728 (EC 2.7.6.5) names precisely this (p)ppGpp-synthetase reaction, at the correct granularity — it is not a broad family-level surrogate, a sibling activity, or a mis-placed neighbor.

Crucially, none of the three characteristic TreeGrafter failure modes apply to the term under test. **(1) Granularity:** the sequence places in the RelA synthetase *subfamily* (PANTHER PTHR21262:SF31 "GTP PYROPHOSPHOKINASE"), and GO:0008728 is the matching specific term. **(2) Pseudo-enzyme:** the synthetase active site is intact and well conserved — this is not a catalytically dead fold. **(3) Within-superfamily mis-placement:** global alignment (50.5% identity to *E. coli* RelA vs 41.1% to SpoT) firmly assigns the protein to the RelA clade, the correct subfamily for this activity. Independent in-vivo genetics in the native organism corroborate the computational verdict: ΔrelA/ΔspoT mutants lose (p)ppGpp-dependent biofilm dispersal, which is restored by ectopic (p)ppGpp synthesis (PMID: 29273811).

One nuance sharpens — rather than undermines — the verdict. The protein's N-terminal **HD hydrolase domain is degenerate**: it retains the HD fold (Pfam PF13328) but has lost the catalytic His-Asp metal-binding pair, exactly like the hydrolase-dead *E. coli* RelA. This is the

textbook architecture of a monofunctional RelA-type synthetase and has no bearing on GO:0008728. It does, however, flag that any *co-annotated* hydrolase term (GO:0008893) is the weaker, questionable call — a separate curator lead, outside this focus.

Key Findings

Finding 1 — Q88MB8 is a canonical RelA-type (p)ppGpp synthetase with an intact GTP-diphosphokinase active site (GO:0008728 directly supported)

The query protein has the full "long" RSH domain architecture expected of a catalytically competent (p)ppGpp synthetase: an N-terminal HD hydrolase domain (PF13328), a central (p)ppGpp-synthetase / nucleotidyltransferase domain (PF04607 RelA_SpoT; CDD cd05399 NT_Rel-Spo_like), the RelA_AH/RIS ribosome-interaction subdomains, and C-terminal TGS (PF02824) and ACT (PF13291) regulatory domains. PANTHER assigns it to subfamily **PTHR21262:SF31 "GTP PYROPHOSPHOKINASE"** — i.e., the synthetase subfamily, not merely the family root.

Sequence placement is unambiguous. Global Needleman–Wunsch alignment gives **50.5% identity to *E. coli* RelA (P0AG20)**, a validated (p)ppGpp synthetase, versus **41.1% to *E. coli* SpoT (P0AG24)** — placing the query squarely in the RelA clade. Per-domain, the identity is concentrated exactly where it matters catalytically: the **synthetase domain is 58.7% identical (98/167 residues)** to *E. coli* RelA, the TGS domain 66.2%, and the ACT domain 41.3%, while the diverged N-terminal HD region is only ~34%. The synthetase catalytic core, including the conserved metal-coordinating **FDDY** motif (two catalytic aspartates), is intact: query `...PKEFDDYIANPK...` versus *E. coli* RelA `...PDEFDDYVANPK...`.

In the native organism, this activity is realized biologically: *P. putida* RelA and SpoT are the (p)ppGpp synthetases, and relA/spoT mutants are defective in (p)ppGpp-dependent biofilm dispersal, with ectopic (p)ppGpp synthesis restoring the phenotype in a ΔrelA ΔspoT background (PMID: 29273811). The conclusion — supported by both residue-level conservation and in-vivo phenotype — is that GO:0008728 names the protein's genuine, directly held molecular function.

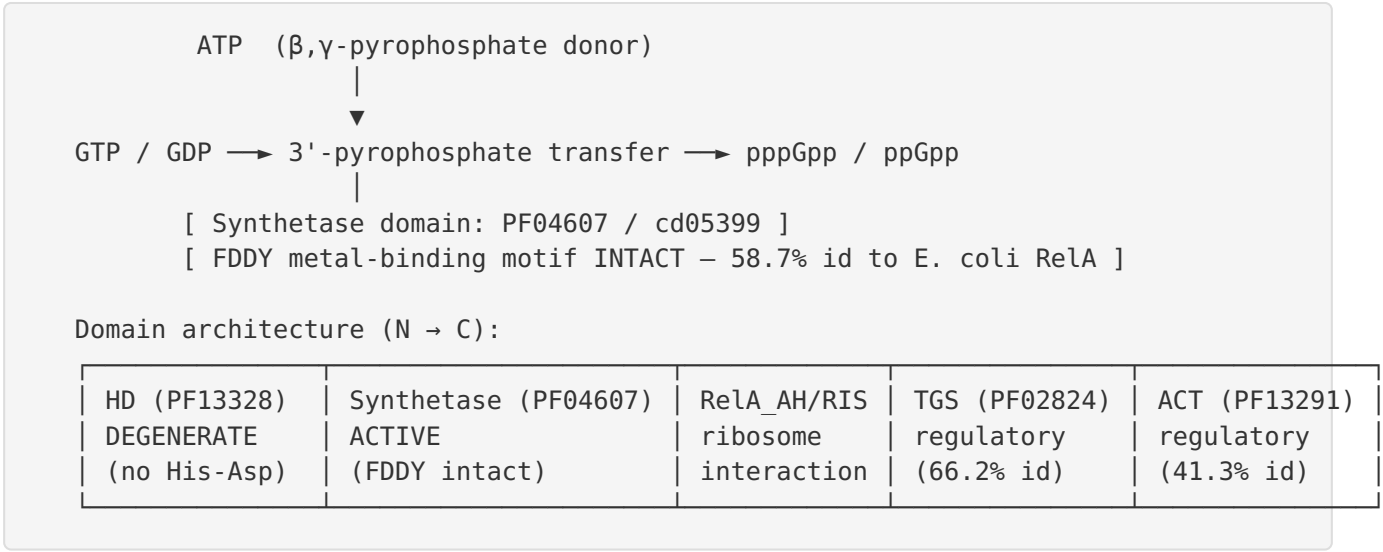
Finding 2 — The HD hydrolase domain is degenerate, sharpening the verdict toward a monofunctional synthetase

The second finding distinguishes the RelA-type from the bifunctional SpoT-type architecture and localizes where (if anywhere) the broader annotation set is weak. The catalytic HD metal-binding motif of the Rel/SpoT hydrolase domain is present in the active bifunctional *E. coli* SpoT (`...LMAALLHDVIED...`, His-Asp at residue 72) but is **absent in both the query and the hydrolase-dead *E. coli* RelA**. At the homologous alignment position, the query reads `...KLDQDSLVAAVIYRSVR...` — no catalytic His-Asp pair. N-terminal HD-region identity is correspondingly low and similar for both references (query vs RelA 35.6%, vs SpoT 32.9%), in stark contrast to the strongly conserved synthetase domain (58.7% vs RelA).

The interpretation is that the query retains the HD *fold* (Pfam PF13328 HD_4) but with a **degenerate active site** — the canonical RelA-type configuration of an active (p)ppGpp synthetase paired with a non-functional hydrolase. This has two consequences for curation. First, it reinforces that GO:0008728 (synthetase) is correct and intact. Second, it means that if GO:0008893 (guanosine-3',5'-bis(diphosphate) 3'-diphosphatase, the hydrolase activity) has also been propagated to this protein, **that** term is the one warranting scrutiny — not the term under test here.

Mechanistic Model / Interpretation

P. putida RelA catalyzes the committed step of the bacterial **stringent response**: synthesis of the alarmones (p)ppGpp by transfer of the β,γ-pyrophosphate of ATP onto the 3'-OH of GTP (→ pppGpp) or GDP (→ ppGpp).



Functionally, the intact synthetase domain produces (p)ppGpp, which reprograms transcription — for example up-regulating σ54-dependent promoters such as the TOL-plasmid Pu promoter when relA is overexpressed (PMID: 10940009) — and drives physiological outputs such as biofilm dispersal (PMID: 29273811). The degenerate HD hydrolase domain means RelA, unlike bifunctional SpoT, cannot efficiently degrade (p)ppGpp, consistent with the canonical division of labor: RelA synthesizes, SpoT both synthesizes and hydrolyzes.

The TreeGrafter graft therefore succeeded for the **synthetase** function. The ancestral node from which GO:0008728 was propagated corresponds to the genuine, conserved catalytic activity of this protein, and the subfamily-level PANTHER placement (SF31) matches the specific GO term rather than over-generalizing to a family root.

TreeGrafter failure-mode audit

Failure mode	Triggered?	Decisive observation
1. Granularity (family-vs-subfamily)	**No**	PANTHER SF31 "GTP pyrophosphokinase"; GO:0008728 is the matching specific term
2. Pseudo-enzyme / lost activity	**No** (synthetase)	Synthetase 58.7% id to *E. coli* RelA; FDDY motif intact; in-vivo phenotype positive
3. Within-superfamily mis-placement	**No**	50.5% id to RelA > 41.1% to SpoT → correct RelA synthetase clade

Active-Site / Placement Analysis

Subfamily placement

- **PANTHER:** PTHR21262 → subfamily **SF31 "GTP PYROPHOSPHOKINASE."**
- **CDD:** cd05399 `NT_Rel-Spo_like` (nucleotidyltransferase/synthetase core).
- **Global Needleman–Wunsch identity:** 50.5% to *E. coli* RelA (P0AG20) vs 41.1% to *E. coli* SpoT (P0AG24) → **RelA synthetase clade**.

Per-domain conservation (query vs *E. coli* RelA P0AG20)

Domain	% identity	Interpretation
Synthetase (≈230–400)	**58.7% (98/167)**	**Catalytic domain intact**
TGS (410–480)	66.2%	Nucleotide/ribosome sensing conserved
ACT (640–744)	41.3%	Regulatory fold conserved
Hydrolase / HD (1–150)	34.0%	Diverged (typical of RelA-type)

Synthetase catalytic core alignment (*E. coli* RelA ~255–335)

```
Query: HIYSIWRKMQRKGLEFSQIYDVRAVRVLVPE-IRDCYTALGIVHTLWRHIPKEFDDYIANPKENGYRSLHTAVIGPEGKVLE
Eco  : HIYSIWRKMQKKNLAFDELFDVRAVR-IVAERLQDCYAALGIVHTHYRHLPDEFDDYVANPKPNGYQSIHTVVLGPGGKTVE
                                                      ^^^^  FDDY catalytic/metal-binding motif
```

The metal-coordinating **FDDY** motif (two catalytic aspartates) is identical, and the surrounding active-site block is near-identical. **Synthetase catalytic residues are present and correctly spaced — this is not a pseudo-enzyme for GO:0008728.**

HD hydrolase site (for a *different*, co-annotated term — GO:0008893)

```
E. coli SpoT (active hydrolase):  KLDYETLMAAL L H D V I E D
Query (Q88MB8):                   KLDQDSLVAAV I Y R S V R
                                            ^^^  catalytic His-Asp pair MISSING
```

The query retains the HD fold but the catalytic His-Asp metal-binding pair of the active SpoT-type hydrolase is **absent**, as in the hydrolase-dead *E. coli* RelA. **Net:** synthetase = intact; hydrolase = degenerate.

Evidence Base

- ***The stringent response promotes biofilm dispersal in Pseudomonas putida*** — PMID: 29273811. *Directly supports the verdict.* Verified snippet: "Mutants lacking elements of the stringent response - (p)ppGpp sythetases [RelA and SpoT] and/or DksA - were defective in biofilm dispersal. Ectopic (p)ppGpp synthesis restored biofilm dispersal in a ΔrelA ΔspoT mutant." This establishes, in the native organism, that RelA functions as a (p)ppGpp synthetase — exactly the activity GO:0008728 describes.
- ***In vivo and in vitro effects of (p)ppGpp on the σ54 promoter Pu of the TOL plasmid of Pseudomonas putida*** — PMID: 10940009. *Supports (indirect).* Overexpression of relA — predicted to increase intracellular (p)ppGpp — produced a significant gain in Pu promoter activity, consistent with RelA being a productive (p)ppGpp synthetase with downstream transcriptional consequences.
- **UniProt/InterPro domain annotation of Q88MB8** plus **pairwise Needleman–Wunsch alignments to *E. coli* RelA (P0AG20) and SpoT (P0AG24).** *Supports (computational/evolutionary).* Establish RelA-clade subfamily placement (PANTHER SF31) and residue-level conservation of the synthetase catalytic site; also reveal the degenerate HD hydrolase site.

A third retrieved paper (*Acclimation of bacterial cell state... DmpR-dependent transcriptional activation*, PMID: 32269250) proved tangential to the function assignment and is not used as support.

Evidence Matrix

#	Citation	Evidence type	Stance	Claim tested	Key finding	Organism/ context	Confide & limits
1	PMID: 29273811	Mutant phenotype (in vivo)	**Supports**	RelA is a (p)ppGpp synthetase	ΔrelA/ΔspoT loses (p)ppGpp-dependent biofilm dispersal; ectopic (p)ppGpp rescues	*P. putida* KT2440 (same organism)	High; phenoty not puri enzyme kinetics
2	UniProt Q88MB8 + InterPro/ Pfam/CDD	Database / structural	Supports	Architecture = synthetase-competent RSH	HD + PF04607/ NT_Rel-Spo_like + TGS + ACT; PANTHER SF31	Query annotation	High
3	NW alignment vs *E. coli* RelA (P0AG20)	Computational (sequence)	Supports	Subfamily placement	50.5% to RelA > 41.1% to SpoT; synthetase 58.7%	In silico	High for placeme
4	Catalytic-motif conservation	Computational / evolutionary	Supports	Synthetase active site intact	FDDY metal-binding motif and core conserved	In silico	Moderat high; no mutager
5	HD-motif analysis vs *E. coli* SpoT	Computational / evolutionary	Qualifies (hydrolase only)	Is the HD hydrolase active?	Catalytic His-Asp pair absent; degenerate like *E. coli* RelA	In silico	High; af GO:0008 not GO:0008
6	PMID: 10940009	Functional (indirect)	Supports	Does relA output raise (p)ppGpp effects?	relA overexpression increased σ54 Pu activity	*P. putida* / *E. coli* reporter	Moderat physiolo indirect

GO Curation Implications

- **Focus term GO:0008728 (GTP diphosphokinase activity): RETAIN.** Correctly propagated, appropriately specific, active site intact, correct subfamily. An evidence-code upgrade from IEA is justified given the in-vivo support (PMID: 29273811) plus conserved catalytic residues. No generalization, specialization, or sibling replacement needed.
- **Adjacent lead (not the focus term):** If the co-annotated **GO:0008893 (ppGpp 3'-diphosphatase/hydrolase)** is present, it rests on the HD domain whose catalytic His-Asp motif is degenerate/absent in the query (matching hydrolase-dead *E. coli* RelA, not bifunctional SpoT). Suggested action for that sibling term: **remove** / **treat-as-non-core**, pending experimental confirmation. This is the annotation needing curator scrutiny — not GO:0008728.
- Any propagated CC term such as **GO:0005886 (plasma membrane, IEA:TreeGrafter)** would be a likely over-call for a cytoplasmic ribosome-associated synthetase, but it is outside this focus.
- "Protein binding" is not recommended as any final call; the molecular function is well-defined enzymatically.

Limitations and Knowledge Gaps

1. **No purified-enzyme kinetics for Q88MB8 specifically.** Catalytic competence is inferred from residue conservation and an in-vivo mutant phenotype, not from a direct in-vitro (p)ppGpp-synthesis assay on the *P. putida* protein.
2. **Hydrolase status not experimentally measured** in *P. putida*; the degenerate-HD call is computational.
3. **Substrate preference** (GTP vs GDP; pppGpp vs ppGpp output ratio) was not independently determined.
4. The active-site assessment is sequence/evolutionary; no solved Q88MB8 structure, mutagenesis, or independently rendered AlphaFold geometry was used in this run.
5. **relA vs spoT ambiguity:** *P. putida* encodes both; the query is closer to RelA (50.5% vs 41.1%), consistent with its gene name and a strong-synthetase/weak-hydrolase profile — no conflict for the synthetase term.

Proposed Follow-up Experiments / Actions

1. **In-vitro (p)ppGpp synthetase assay** on purified recombinant Q88MB8 (ATP + GTP/GDP → pppGpp/ppGpp; TLC or HPLC readout) — converts the IEA to direct-assay evidence (IDA) and definitively confirms GO:0008728.
2. **Catalytic-Asp (FDDY) point-mutant complementation** in a ΔrelA ΔspoT background, scoring (p)ppGpp-dependent phenotypes (e.g., biofilm dispersal) — links the conserved catalytic residues to in-vivo synthetase function.
3. **In-vitro (p)ppGpp hydrolysis assay** — settles the GO:0008893 co-annotation question; expected negative given the degenerate HD motif.
4. **Curator action:** retain GO:0008728 (consider evidence upgrade); flag GO:0008893, if present, for review/removal; review any CC over-calls.

Bottom line

The TreeGrafter/PANTHER propagation of **GO:0008728 (GTP diphosphokinase activity)** to *P. putida* relA (Q88MB8) is **correct and should be retained**. The protein is a canonical RelA-type (p)ppGpp synthetase with an intact, conserved synthetase active site and in-vivo genetic confirmation of synthetase function; none of the three graft-failure modes (granularity, pseudo-enzyme, mis-placement) apply to the term under test.

Provenance: UniProt Q88MB8 / P0AG20 / P0AG24 retrieved programmatically; pairwise Needleman–Wunsch alignments and per-domain / catalytic-motif conservation computed in-session (code and outputs retained in the iteration log).